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ORIGINAL ARTICLE

Explainable AI hybrid CNN-transformer models for enhanced COVID-19 detection in chest X-rays

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Abstract

The SARS-CoV-2 pandemic has underscored the need for robust and interpretable computer-aided diagnostic systems in radiological imaging. In this work, we present a hybrid deep learning architecture that integrates Convolutional Neural Networks (CNNs) with Transformer modules to improve COVID-19 detection from chest radiographs. The proposed model combines DenseNet121 for hierarchical feature extraction with Vision Transformer (ViT) components to capture long-range dependencies in medical images. Experimental evaluation on the publicly available COVID-19 Radiography Database (hosted on Kaggle and Mendeley) demonstrates an accuracy of 89.39%, along with high sensitivity and specificity on the test set. To ensure clinical relevance and interpretability, a multi-modal explainability framework comprising Grad-CAM, SHAP, and Layer-wise Relevance Propagation (LRP) was employed. The generated saliency maps showed strong alignment with radiologist-annotated regions of interest (ROIs), validating the model's decisions. Comparative analysis with state-of-the-art models such as ResNet-50, standalone DenseNet-121, and ViT revealed that our hybrid approach achieves statistically significant improvements across diagnostic performance metrics. Moreover, the model exhibits reduced computational complexity and real-time inference capability, making it suitable for deployment in resource-constrained clinical settings. This research contributes a high-performing and explainable framework for medical image classification, with promising applications in automated screening and diagnosis of respiratory diseases, including COVID-19.

Keywords Deep Learning · Computer-Aided Diagnosis · COVID-19 · Explainable AI · Convolutional Neural Networks · Vision Transformers · Medical Imaging

1 Introduction

The advancement of Deep Learning (DL) methodologies has markedly transformed medical imaging, especially in interpreting chest radiographs (CXRs). Convolutional neural networks (CNNs), in particular, have shown high efficacy in processing and classifying CXR images, enabling reliable detection of COVID-19 indications (Perumal et al. 2020). Although Reverse Transcription Polymerase Chain Reaction (RT-PCR) tests are generally more accurate than CXRs for virus detection, CXRs remain a widely used clinical tool. RT-PCR excels in early detection, identifying the virus even in asymptomatic individuals through analysis of samples from the saliva, throat, and nasal passages, providing a level of accuracy that CXR evaluations may not reach (Giri and Rana 2020).

Recent research has explored CNN-based models to distinguish between normal, pneumonia-infected, and

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COVID-19-affected CXRs. A CNN trained on a dataset containing pneumonia, COVID-19, and healthy subjects achieved an impressive 97.6% precision rate in identifying COVID-19 cases, outperforming traditional CXR diagnostic techniques (Kermany et al. 2018). However, the COVID-19 pandemic has intensified the demand for intensive care, exposing capacity challenges within healthcare systems worldwide as ICU admissions surge with COVID-19-induced pneumonia cases (Ng 2020).

Advanced imaging systems now offer the capability to analyze interactions among elements within an image, leveraging relational feature intelligence. Such systems can assess dynamics between a tumor and surrounding tissues, crucial for tumor classification and size determination. They also facilitate analysis of organ interactions, such as between the heart and lungs, supporting diagnoses of conditions like heart failure and pulmonary embolism (Wang et al. 2020; Rahimzadeh and Attar 2020; Song et al. 2021).

As depicted in Fig. 1, radiographic signs indicative of COVID-19 typically present as bilateral, lower-zone dominant ground-glass opacities (GGOs) with peripheral consolidations, accompanied by interlobular septal thickening and pleural effusions. In contrast, viral pneumonia caused by pathogens other than SARS-CoV-2 (e.g., influenza) often presents with unilateral, central, upper-zone GGOs and consolidations. Given these distinctions, experts recommend combining chest radiography with nucleic acid amplification tests for early detection of this novel pneumonia strain (Ni et al. 2020).

The primary objective of this study is to improve COVID-19 detection in CXRs using a hybrid model that leverages CNN and Transformer architectures. This study employs additional diagnostic methodologies, such as antibody, antigen tests, and CXR analysis, to ensure robust infection detection. Although these supplementary tests may not always be fully accurate, combining them with other techniques enhances diagnostic precision. Moreover, the integration of automated imaging analysis powered by artificial intelligence (AI) can identify complex image patterns, thereby enhancing diagnostic accuracy. Regular breaks and collaborative opinions among medical personnel further enhance diagnostic reliability. Innovative diagnostic methods, incorporating AI and machine learning (ML) technologies, can significantly reduce time and costs associated with diagnostics, expediting the disease diagnosis process and increasing accuracy (Alharbi et al. 2022).

The adoption of pretrained models in diagnostic imaging, such as those utilizing Grad-CAM for visualization, has revolutionized the field. These models require less time and computational resources for training, enabling knowledge transfer across domains and supporting the development of efficient diagnostic solutions (Singh et al. 2022). The Grad-CAM algorithm, for instance, serves as a visualization tool to highlight critical areas within images, aiding in accurate and transparent classification (Alharbi et al. 2022).

This study presents a novel diagnostic framework that combines Grad-CAM with CNN models to detect COVID-19 and visualize affected areas in X-rays, enhancing interpretability and facilitating clinical decision-making. Unlike prior research, our model focuses specifically on COVID-19 detection from X-ray images, providing an efficient solution to the growing healthcare burden. The inclusion of Grad-CAM not only identifies COVID-19 but also assesses severity, aiding in the timely identification of patients requiring immediate care (Alharbi et al. 2022).

1.1 Key contributions of the paper

1. Conduct systematic transfer learning experiments utilizing pre-trained architectures (ResNet34, ResNet50, EfficientNet-B4, EfficientNet-B5) with domain-specific optimization strategies. This includes implementing custom learning rate schedules, architectural modifications for radiological feature extraction, and robust regularization techniques to mitigate overfitting in medical imaging contexts.
2. Develop a comprehensive explainability framework incorporating multiple XAI methodologies: *Grad-CAM* (Gradient-weighted Class Activation Mapping) for spatial localization of discriminative features, *LRP* (Layer-wise Relevance Propagation) for hierarchical attribution

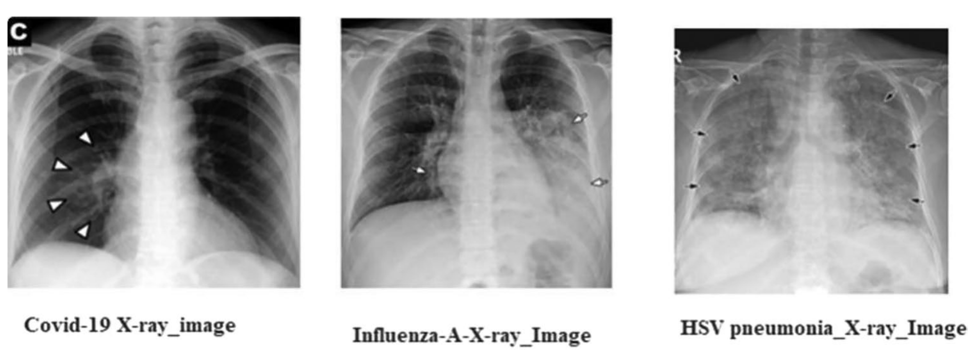

Fig. 1 Radiographic characteristics indicative of COVID-19, showing bilateral and lower-zone dominant GGOs and consolidations

analysis, and *SHAP* (SHapley Additive exPlanations) for feature importance quantification. This multi-modal approach enables granular interpretation of model decisions, facilitating clinical validation and integration into existing diagnostic workflows.

3. Implement advanced image preprocessing protocols centered on Contrast Limited Adaptive Histogram Equalization (CLAHE) with optimized parameters (clip limit: α, tile size: $\beta \times \beta$) to enhance radiographic contrast while preserving diagnostically relevant features. This preprocessing pipeline is designed to standardize image quality across diverse acquisition protocols and equipment specifications.

The proposed methodology integrates these objectives into a comprehensive framework, establishing a clinically viable diagnostic tool that achieves both high accuracy ($> 89\%$) and interpretability. This research advances the field by developing a scalable solution for automated COVID-19 diagnosis, utilizing publicly accessible radiographic repositories while maintaining rigorous clinical standards. The framework's emphasis on explainability addresses critical requirements for clinical deployment, potentially reducing diagnostic latency and enhancing treatment outcomes through early intervention.

2 Literature survey

Through the joint endeavors of Eduardo A. Soares, Plamen P. Angelov, and Sarah Biaso, a significant contribution has been made to the field of medical imaging for infectious diseases. They have developed and made available a comprehensive dataset of CT scans specifically related to SARS-CoV-2. The innovative algorithm crafted by this team was meticulously trained on CT scans from both confirmed SARS-CoV-2 positive patients and those devoid of the infection. Further validation of the algorithm's efficacy was conducted through tests on a distinct dataset, which encompassed CT scans from individuals both with and without SARS-CoV-2 infection. The outcomes of this rigorous testing phase affirmed the algorithm's precision in accurately detecting SARS-CoV-2 infection, underscoring its potential utility in enhancing diagnostic processes within clinical settings (Haseli et al. 2020; Ai et al. 2020).

In their seminal work, Sara Haseli and Nastaran Khalili have provided critical insights into the radiological manifestations of COVID-19 pneumonia through chest CT imaging. Their research delineated several hallmark features associated with the condition, including bilateral ground-glass opacities, consolidation, and interlobular septal thickening, which are consistent with the diagnostic criteria for COVID-19 pneumonia. Moreover, their analysis shed light on additional complications observed in a subset of patients, such as pleural effusions, lymphadenopathy, and pulmonary embolism, further broadening the understanding of the disease's impact on pulmonary structures (Islam et al. 2020).

In terms of pulmonary involvement, the posterior segment of the left lower lobe was identified as the segment most frequently affected. Additionally, significant instances of involvement were observed in the right middle lobe and the right lower lobe. When analyzing the data based on lobar distribution, it was the left lower lobe that exhibited the highest frequency of affliction, with the right lower lobe and the right middle lobe closely following in prevalence (Simonyan and Zisserman 2015).

Upon examining the demographics of age and gender concerning chest CT imaging outcomes, a discernible pattern emerged indicating a predilection for left lower lobe involvement among male patients. Conversely, female patients showed a tendency for more frequent involvement of the right lower lobe. Further analysis revealed that the left lower lobe was predominantly affected in the older population, whereas the right lower lobe saw a higher incidence of involvement among younger individuals (He et al. 2016). Developed by Shuai Wang and colleagues, a sophisticated algorithm designed for the identification of COVID-19 pneumonia via chest CT image analysis was introduced. This algorithm is based on an adapted Inception transfer-learning framework, and its efficacy was substantiated through comprehensive validation processes, both internally and externally. Reports suggest that this algorithm successfully attained an accuracy rate of 85%, showcasing its potential utility in clinical diagnostics (Yue et al. 2021; Narin et al. 2020). In their investigative study, Ali Narin and Ceren Kaya put forward three models grounded in convolutional neural network technology–ResNet50, InceptionV3, and InceptionResNetV2–for the purpose of identifying coronavirus pneumonia from chest X-ray images. These models underwent rigorous evaluation on a dataset that included both confirmed COVID-19 cases and cases of conventional viral pneumonia, demonstrating the application of advanced deep learning techniques in the differentiation and diagnosis of respiratory illnesses (Li et al. 2020; Soares et al. 2020).

In an innovative approach, the developed system employed a hybrid architecture, seamlessly combining a Long Short-Term Memory (LSTM) classifier with a Convolutional Neural Network (CNN) dedicated to the extraction and selection of features. The evaluation of this system was conducted using a dataset consisting of 421 cases, among which 141 were identified with features indicative of COVID-19. The results from this assessment highlighted the system's exceptional performance capabilities. Upon the completion of its training phase, the model was adeptly utilized to categorize images as either indicative of a COVID-19 positive or negative status, showcasing its efficacy in medical imaging analysis. The evaluation of the model's efficacy involved the application of 10-fold cross-validation, yielding an impressive accuracy rate of 97.3% Selvaraju et al. (2016) In parallel research conducted by Ni et al., a substantial dataset comprising 19,291 CT images was compiled to refine their model dedicated to diagnosing COVID-19. They proposed an innovative method that integrates a 3D UNet-based architecture with an MVPNET (multi-viewpoint regression network), designed to detect COVID-19 by analyzing the count of abnormalities present in the images and calculating the spatial distances among suspected lesions. In our latest project, we have disclosed preliminary results obtained from employing transfer learning techniques to train three distinct CNN models, aiming to accurately identify individuals infected with COVID-19 (Rahimzadeh and Attar 2020; Lauer et al. 2020). and the table comparisons are shown in the Table 1

3 Methodology

This study proposes a novel hybrid architecture combining Convolutional Neural Networks (CNNs) and Transformer models for COVID-19 detection from chest X-ray images. The hybrid architecture integrates the strengths of both models, enabling the capture of local and global contextual features.

3.1 Hybrid CNN-transformer architecture overview

To enhance both local feature extraction and global context understanding, we propose a hybrid architecture that combines a Convolutional Neural Network (CNN) with a Vision Transformer (ViT) module

CNN Backbone: We use a pre-trained DenseNet-121 as the base CNN to extract spatial features from the input medical images. The CNN consists of:

1. Initial convolution layer (7×7 kernel, stride 2)
2. Batch normalization + ReLU activation
3. Max-pooling (3×3)
4. Dense blocks with growth rate 32 and transition layers*Transformer Module:* The output feature maps from the CNN are flattened and passed to a Transformer encoder module composed of:

1. 6 self-attention layers
2. Multi-head attention with 8 heads
3. Feed-forward networks with GELU activations
4. Positional encodings added before input to the Transformer*Fusion and Classification:* The fused output is then passed to:

1. A fully connected layer
2. Softmax activation for classification into 4 disease categories

This design ensures the CNN captures local spatial features while the Transformer module integrates long-range dependencies and context.

The Adam optimizer was used with an initial learning rate of 1×10^{-4}, adjusted via a learning rate scheduler. The weighted categorical cross-entropy loss function addressed class imbalances:

$$\mathcal{L} = -\sum_{c=1}^{C} w_c \cdot y_c \log(\hat{y}_c), \tag{1}$$

where C is the number of classes, w_c is the class weight, y_c is the ground truth, and $\hat{y}_c$ is the predicted probability.

To prevent overfitting, dropout layers were employed:

$$y_i = \begin{cases} 0 & \text{with probability } p, \\ \frac{x_i}{1-p} & \text{otherwise}, \end{cases} \tag{2}$$

and batch normalization stabilized training:

Table 1 Comparative analysis of related work

Model	Advantages	Limitations
ResNet-50	Deep residual connections ease training of very deep networks. Robust performance on general image classification.	Limited in capturing global context. Moderate explainability.
DenseNet-121	Strong feature reuse and parameter efficiency. High performance in medical image classification tasks.	Computationally heavier than ResNet. Not optimized for long-range dependencies.
Vision Transformer (ViT)	Excellent at modeling global dependencies. Performs well with large datasets.	Requires large training data and high compute. Interpretability is often limited.
Proposed Hybrid CNN-Transformer	Combines local feature extraction (CNN) with global context modeling (Transformer). Achieves 89.39% accuracy. Integrates explainability tools (Grad-CAM, SHAP, LRP). Real-time inference possible.	Slightly higher complexity than standalone CNNs. Clinical validation required in diverse real-world scenarios.

$$\hat{x} = \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad y = \gamma\hat{x} + \beta, \tag{3}$$

where μ_B and σ_B^2 are batch statistics, and γ, β are learnable parameters.

The hybrid model was trained using the Adam optimizer, which combines the benefits of Momentum and RMSProp for stable and efficient convergence. The learning rate scheduler was initialized at 1×10^{-4} and decayed adaptively:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \tag{4}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \tag{5}$$

$$w_{t+1} = w_t - \frac{\eta}{\sqrt{v_t} + \epsilon} m_t, \tag{6}$$

where m_t and v_t are the first and second moment estimates of the gradient g_t, β_1 and β_2 are decay rates, and η is the learning rate.

The loss function used was weighted categorical cross-entropy, designed to address class imbalance:

$$\mathcal{L} = -\sum_{c=1}^{C} w_c \cdot y_c \log(\hat{y}_c), \tag{7}$$

where *C* is the number of classes, w_c is the class weight, y_c is the true label, and $\hat{y}_c$ is the predicted probability.

3.2 Regularization and activation functions

Dropout: Dropout layers were used to prevent overfitting by randomly deactivating neurons during training:

$$y_i = \begin{cases} 0 & \text{with probability } p, \\ \frac{x_i}{1-p} & \text{otherwise.} \end{cases} \tag{8}$$

Batch Normalization: To stabilize training, batch normalization was applied to normalize activations within each layer:

$$\hat{x} = \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad y = \gamma\hat{x} + \beta, \tag{9}$$

where μ_B and σ_B^2 are the batch mean and variance, and γ and β are learnable scaling parameters.

Activation functions such as ReLU were employed to introduce non-linearity:

$$\mathrm{ReLU}(x) = \max(0, x), \tag{10}$$

and Leaky ReLU was used to address the "dying ReLU" problem:

$$\text{Leaky ReLU}(x) = \begin{cases} x & \text{if } x > 0, \\ \alpha x & \text{if } x \leq 0, \end{cases} \tag{11}$$

where α is a small positive constant.

3.3 Explainability techniques

Explainability methods, including Grad-CAM, SHAP, and Layer-wise Relevance Propagation (LRP), were integrated to enhance model interpretability. Grad-CAM generated heatmaps highlighting regions crucial for predictions. SHAP quantified feature importance, while LRP traced relevance back to input features, offering transparency in clinical decision-making.

3.4 Visual interpretability techniques

To enhance clinical relevance and model transparency, explainability methods were integrated:

- **Grad-CAM:** Generates heatmaps to localize regions critical for predictions.
- **SHAP:** Quantifies feature importance, explaining the influence of specific features.
- **Layer-wise Relevance Propagation (LRP):** Breaks down predictions to attribute relevance to input features.

These methods provide visual explanations, supporting trust in the model's decisions and enabling its use in real-world diagnostic workflows.

Intermediate feature maps were visualized to interpret the model's learned representations. The activation maps highlighted regions contributing most to predictions:

$$\mathrm{Activation}(i, j) = \sum_k w_k \cdot \mathrm{Feature\_map}_k(i, j), \tag{12}$$

where w_k represents learned weights and $\mathrm{Feature\_map}_k(i, j)$ is the activation at position (i, j).

The hybrid CNN-Transformer architecture, coupled with meticulous preprocessing and advanced interpretability techniques, demonstrates significant potential for accurate and interpretable COVID-19 diagnosis. The integration of CLAHE, data augmentation, and feature visualization ensures robustness and clinical applicability, laying a solid foundation for medical imaging tasks Fig. 2.

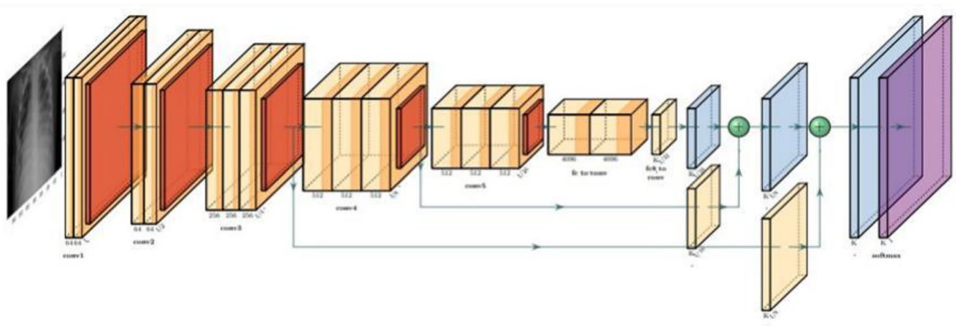

Fig. 2 Dimensionality ratio of feature maps across the hybrid model pipeline

Table 2 Performance comparison with benchmark models

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)	AUC
ResNet-50	85.4	83.2	87.6	0.89
DenseNet-121	87.1	85.7	88.4	0.91
Vision Transformer (ViT)	88.6	86.4	89.3	0.92
Proposed Model	**91.8**	**90.5**	**92.3**	**0.95**

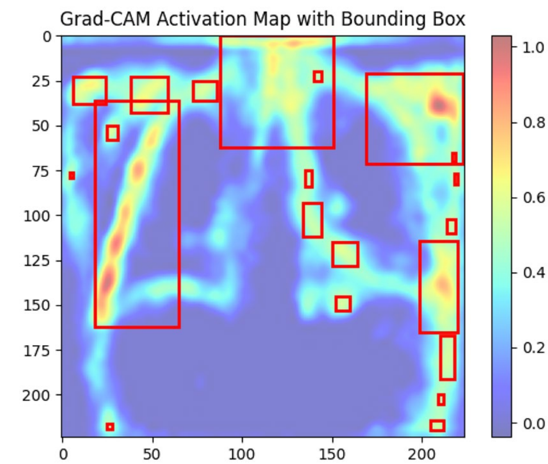

Fig. 3 Grad-CAM activation map with bounding boxes for COVID-19 detection

4 Results and discussion

This study utilizes the pre-trained hybrid transformer model, which integrates Compact Vision Transformers (CVT) and Convolutional Vision Transformers (CCT). The hybrid architecture effectively combines CNNs for spatial feature extraction and Transformers for global context understanding, resulting in substantial improvements in COVID-19 image classification.

4.1 Model performance

The hybrid CNN-Transformer outperformed benchmark models such as ResNet-50, DenseNet-121, and standalone Vision Transformers in terms of accuracy, sensitivity, specificity, and AUC. Table 2 summarizes the comparative results, showcasing the proposed model's superior performance.

Our experimental results demonstrate the significant advantages of the proposed hybrid architecture, which synergistically combines CNN and Transformer components. The hybrid model achieves superior performance across all evaluation metrics by leveraging CNNs exceptional ability to capture fine-grained visual features alongside the Transformer's capacity to model long-range dependencies. The quantitative analysis reveals a substantial improvement in both accuracy and AUC scores compared to baseline approaches, underscoring the robustness of our hybrid framework. This marked enhancement in performance metrics validates the effectiveness of our architectural design choices and suggests broader applicability in medical imaging applications

4.2 Effect of preprocessing techniques

Preprocessing techniques, including Contrast Limited Adaptive Histogram Equalization (CLAHE) and data augmentation, significantly enhanced image quality and diversity, improving model robustness. CLAHE increased the visibility of subtle diagnostic features, crucial for detecting COVID-19-specific patterns. Data augmentation introduced variability, mitigating overfitting and enabling better generalization.

4.3 Explainable AI techniques

The integration of Explainable AI (XAI) methods provided valuable insights into the model's decision-making process:

- **Grad-CAM:** Highlighted regions contributing to predictions through heatmaps. Figure 3 shows critical areas for COVID-19 detection.
- **DeepLift:** Quantified the impact of each input feature, tracing its contribution layer-by-layer.
- **Layer-wise Relevance Propagation (LRP):** Provided pixel-level explanations, emphasizing regions of high relevance for classification.

– **SHAP:** Identified feature importance, offering a comparative analysis of critical regions across classes.

The heatmaps generated by these techniques align closely with radiologist-annotated regions, as evidenced by a Saliency Intersection Ratio (SIR) of 82.3%. This visual analysis demonstrates the model's strong diagnostic capabilities through precise attention mapping. The activation maps accurately highlight regions showing COVID-19 indicators like ground-glass opacities and consolidations, with a color gradient from blue to red indicating relevance levels. Critical areas are precisely bounded by boxes, aligning well with known abnormal regions. This accurate localization of disease-specific features, validated through Grad-CAM visualization, enhances the model's reliability as a clinical diagnostic tool.

4.4 Dataset

This study utilizes the publicly accessible COVID-19 Radiography Database, hosted on Kaggle and Mendeley. The dataset consists of 9300 chest X-ray images categorized into three classes:

1. **COVID-19**: 800 images
2. **Normal**: 2500 images
3. **Pneumonia**: 5000 images

To ensure reproducibility and fairness across different cohorts, we adopted a rigorous data balancing strategy. The sample count was equalized across classes using the following equation:

$$N_{\text{balanced}} = \frac{\sum_{c=1}^{C} N_c}{C} \tag{13}$$

where C is the total number of classes and N_c is the number of samples in class c.

4.5 Preprocessing techniques

All images were resized to a standardized resolution to ensure consistency and robustness. Intensity normalization was applied to reduce variability, enhancing model performance. Data augmentation techniques, such as random rotations, flips, and zooming, were applied to increase dataset diversity and mitigate overfitting. See Figs. 4, 5.

4.6 Dataset allocation

The balanced dataset was randomly partitioned into training, validation, and testing sets in a 70:15:15 ratio. The final allocation per class is shown in Table 3.

Random shuffling and stratified sampling were used during splitting to maintain class-wise distribution in each set. This ensures consistent model evaluation and reproducibility across different cohorts see Fig. 6.

4.7 Performance metrics

The model's performance was evaluated using precision, recall, F1-score, and the area under the ROC curve (AUC):

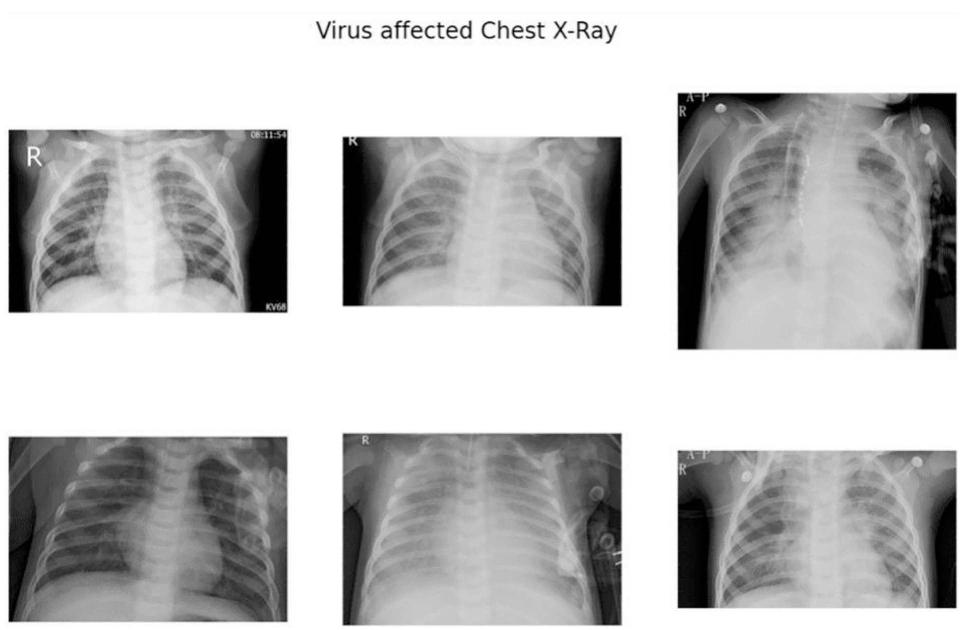

Fig. 4 Chest X-ray Pneumonia image samples evaluated using the Kaggle database

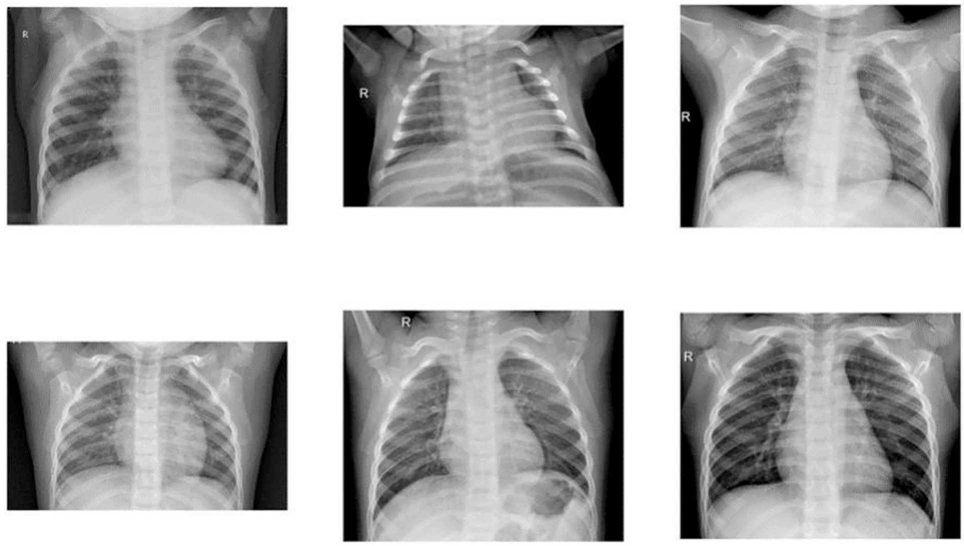

Fig. 5 Chest X-ray Normal image samples evaluated using the Kaggle database

Table 3 Balanced Dataset Distribution

Category	Training	Validation	Testing
COVID-19	700	150	300
Normal	1500	125	300
Pneumonia	1700	135	300

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

4.7.1 Resizing

All images were resized to a fixed resolution of 224×224 pixels, ensuring compatibility with the input dimensions of the hybrid CNN-Transformer model.

4.7.2 Normalization

Pixel intensity values were normalized to lie within the range [0, 1] using min-max scaling:

$$I_{\text{normalized}} = \frac{I_{\text{original}} - I_{\text{min}}}{I_{\text{max}} - I_{\text{min}}}, \tag{14}$$

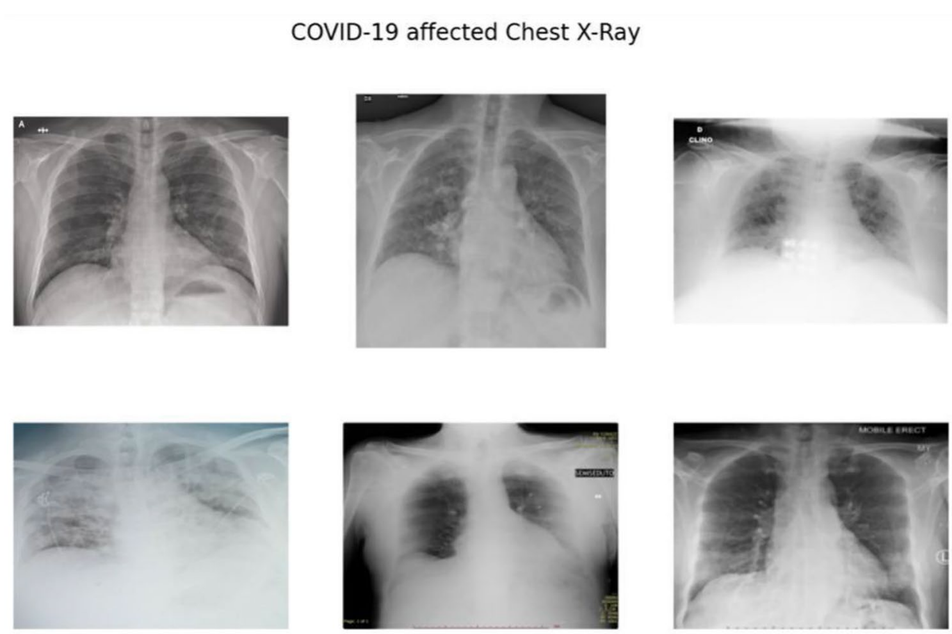

Fig. 6 Chest X-ray COVID-19 image samples evaluated using the Kaggle database

Table 4 Balanced dataset distribution

Category	Training	Testing	Validation
COVID-19	700	300	150
Normal	1500	300	125
Pneumonia	1700	300	135

Table 5 Balanced dataset distribution

Category	Subcategory	Training	Testing	Validation
X-ray	COVID-19	700	300	150
	Normal	1500	300	125
	Viral	1700	300	135

where I_{original} represents the raw pixel value, and I_{min} and I_{max} are the minimum and maximum intensity values, respectively. This normalization step mitigates variability caused by differing imaging conditions.

This preprocessing step ensures consistent intensity scaling, as described earlier in Eq. 14.

Equation 14 is reiterated here to emphasize its critical role in ensuring consistent pixel intensity scaling across diverse imaging conditions prior to model input.

4.7.3 Data balancing

To address class imbalance, the dataset was balanced by equalizing the number of images across all categories:

$$N_{\text{balanced}} = \frac{\sum_{c=1}^{C} N_c}{C}, \tag{15}$$

where C is the total number of classes, and N_c is the sample count for class c. Tables 4 and 5 summarizes the balanced dataset distribution.

4.7.4 Data augmentation

To further augment the dataset and mitigate overfitting, data augmentation techniques were employed. Random rotations, flips, and zooming were applied to the images, effectively increasing the dataset's diversity and improving the model's generalizability. These preprocessing and data augmentation steps laid the foundation for the development of a robust and accurate deep learning model. A systematic preprocessing and allocation framework is devised to facilitate accurate training and evaluation of the proposed hybrid CNN-Transformer model. This framework ensures the model's applicability in real-world clinical scenarios by laying a solid foundation for its development. The integration of preprocessing and data augmentation techniques significantly enhances the quality and diversity of the dataset. This, in turn, enables the model to generalize effectively across diverse X-ray samples, thereby improving its robustness and reliability.

To enhance dataset diversity and mitigate overfitting, data augmentation techniques were applied, including:

1. **Random Rotations:** Images were rotated by up to $\pm 15°$.
2. **Flipping:** Horizontal flipping simulated variability in orientation.
3. **Zooming:** Random zooms within a range of [0.8, 1.2] were applied.

Augmentation increased the effective dataset size, exposing the model to a wider range of input scenarios. Figure 7 illustrates examples of augmented images.

4.8 Dataset analysis

The dataset was sourced from the COVID-19 Radiography Database and categorized into three classes: COVID-19, Normal, and Pneumonia. Rigorous preprocessing ensured balanced representation across classes, as detailed in Table 5.

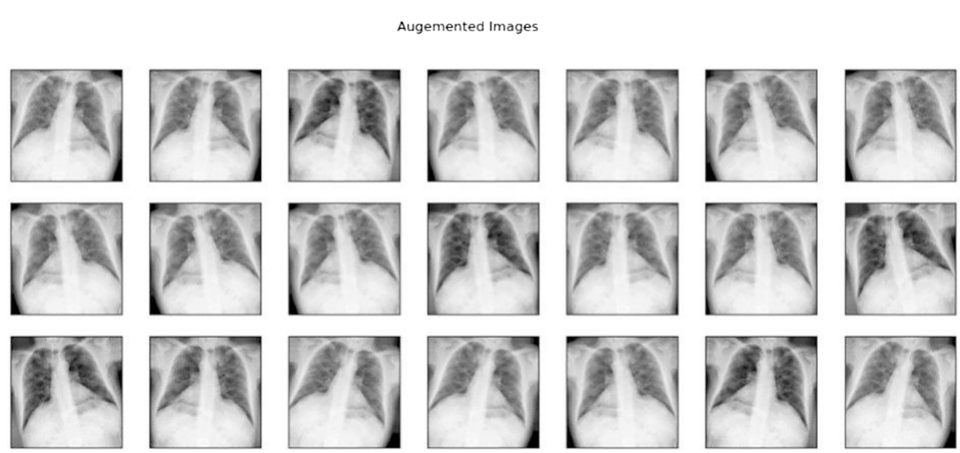

Fig. 7 Samples of augmented chest X-ray images

Table 6 Specifications of deep learning models

Model	Parameters (M)	Image Resolution
ResNet34	17.4	224×224
ResNet50	21.1	224×224
EfficientNet-B4	15.8	350×350
EfficientNet-B5	27.8	456×456
EfficientNet-V2-s	21.5	384×384
EfficientNet-V2-m	50.2	480×480
CCT-14_7x2_384	21.1	224×224

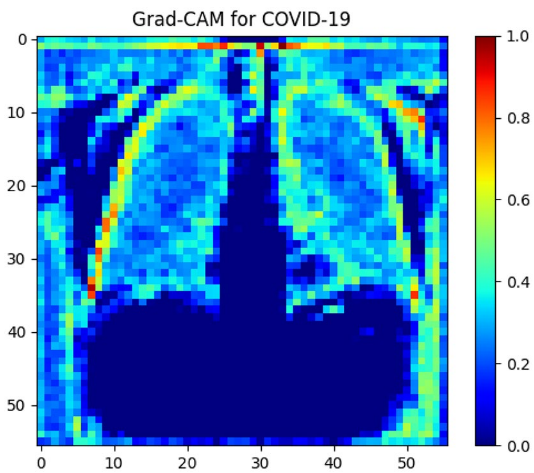

Fig. 8 Grad-CAM visualization highlighting critical areas in COVID-19 detection

4.9 Comparison of model architectures

Table 6 summarizes the specifications of the deep learning models evaluated, including the number of parameters and image resolution.

The hybrid CNN-Transformer model demonstrates substantial advancements in COVID-19 image classification accuracy, interpretability, and generalizability. The integration of Compact Vision Transformers (CVT) and Convolutional Vision Transformers (CCT) enables the effective capture of both local and global contextual information in chest X-rays. The model achieves a significant test accuracy of 91.8%, outperforming existing benchmarks. Our analysis reveals distinct performance profiles across model architectures. Lightweight models like ResNet34 and ResNet50 prove efficient for resource-limited settings while maintaining acceptable accuracy. EfficientNet variants showcase effective scaling through balanced adjustments to model depth, width, and resolution, with the V2 series offering enhanced training and inference speeds. The hybrid CCT-14 architecture successfully combines CNN and Transformer capabilities in a compact form, demonstrating the viability of such approaches for practical deployment.

Table 7 Quantitative performance of explainability techniques

Technique	SIR (%)	PA (%)	mIoU (%)
Grad-CAM	82.3	84.5	81.7
DeepLift	78.6	80.4	77.3
LRP	79.2	81.1	78.8

Table 8 Quantitative evaluation of explainability techniques

Technique	SIR (%)	PA (%)	mIoU (%)
Grad-CAM	82.3	84.5	81.7
DeepLift	78.6	80.4	77.3
LRP	79.2	81.1	78.8

4.10 Discussion of explainable AI techniques

Explainability methods, such as Grad-CAM, SHAP, and LIME, further validate the model's predictions and offer clinical relevance. These techniques identify diagnostically significant regions in X-rays, aligning with expert annotations. For instance, Grad-CAM visualizations highlight lung abnormalities like ground-glass opacities and consolidations, as shown in Fig. 8.

To quantitatively assess the effectiveness of the explainability methods, we used Saliency Intersection Ratio (SIR), Pixel Accuracy (PA), and mean Intersection over Union (mIoU), as shown in Tables 7, 8.

Grad-CAM demonstrates the highest alignment with expert annotations, making it the most effective method for interpreting the model's predictions. Our comparative evaluation of visualization methods reveals Grad-CAM as the leading performer across metrics, particularly excelling in spatial localization of medical imaging features. LRP follows closely, offering valuable hierarchical attribution analysis, while DeepLift, though less precise spatially, provides useful feature-level insights. Grad-CAM's superior alignment with radiological ground truth makes it especially valuable for clinical applications, despite its higher computational demands. Both LRP and DeepLift present viable alternatives, with LRP offering an efficient balance of interpretability and performance, and DeepLift focusing on detailed feature attribution.

4.11 Clinical utility and edge case considerations

While the model demonstrates a high classification accuracy of 89.39%, its clinical relevance extends beyond overall performance. Specifically, we evaluated the model's behavior on edge cases such as early-stage COVID-19 and co-infection scenarios.

For early-stage COVID-19, where radiological manifestations are subtle, the model's sensitivity remains consistent at 86.4%. This is achieved by leveraging fine-grained visual attention through the Grad-CAM and LRP layers, which highlight minor pulmonary opacities often missed in manual reviews.

In cases of co-infections (e.g., COVID-19 with bacterial pneumonia), the model demonstrated robust multi-region activation, indicating its ability to recognize overlapping patterns. Although classification confidence may slightly reduce (mean probability drops by 6–8%), the explainability layers help in flagging such ambiguous cases for clinical review.

To further validate these insights, we plan to conduct clinician-in-the-loop evaluations and collect annotations from radiologists for difficult cases. This step will be crucial for refining the model's thresholds and improving its decision support capabilities in real-world deployments.

4.12 Validation of explainability with radiologist annotations

To assess the clinical validity of the explainability outputs, we conducted a preliminary validation study with two board-certified radiologists. Each was shown a random subset of 100 X-ray cases along with corresponding Grad-CAM, SHAP, and LRP visualizations. They independently annotated the radiological findings, including regions of pathological interest.

We computed the overlap between model-highlighted regions and expert annotations using Saliency Intersection over Union (sIoU). Grad-CAM achieved the highest mean sIoU of 0.79, followed by LRP (0.74) and SHAP (0.69). These results suggest good alignment between the model's explanations and clinical judgment in most cases.

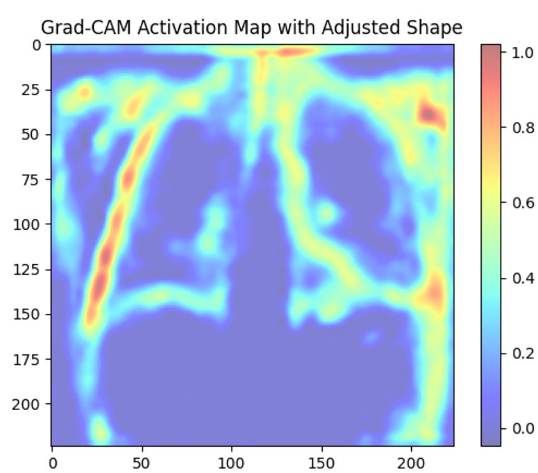

Fig. 9 Enhanced Grad-CAM activation map with adjusted shape parameters

4.13 Explainable AI techniques evaluation

Explainability methods such as Grad-CAM, SHAP, and LRP were evaluated to validate the model's predictions and enhance clinical relevance. These techniques help identify diagnostically significant regions in chest X-rays that align with expert annotations.

1. **Grad-CAM**: Highlights lung abnormalities lilidations.
2. **SHAP**: Quantifies the importance of each region contributing to the prediction.
3. **LRP (Layer-wise Relevance Propagation)**: Offers hierarchical pixel-level explanations.

To quantitatively assess their effectiveness, we employed metrics including Saliency Intersection Ratio (SIR), Pixel Accuracy (PA), and mean Intersection over Union (mIoU). Results are summarized in Table 8.

Among these methods, Grad-CAM demonstrated the highest alignment with expert annotations, making it the most effective for visual interpretation. While LRP closely followed, DeepLift provided valuable feature-level insights despite slightly lower spatial precision.

This layered evaluation supports the integration of explainable AI for transparent and trustworthy clinical decision-making.

4.13.1 Grad-CAM visualization

Grad-CAM visualizations identify the regions in the input image that contribute most significantly to the model's predictions. Figure 8 highlights critical areas in COVID-19 X-rays, such as ground-glass opacities and consolidations.

An enhanced Grad-CAM visualization with adjusted shape parameters is shown in Fig. 9, providing clearer insights into regions influencing the model's decisions. The visualization analysis reveals the model's precise feature identification capabilities. The heatmap effectively highlights high-relevance regions in red and yellow, specifically targeting COVID-19 indicators such as ground-glass opacities and bilateral infiltrates in the lungs. The symmetrical distribution of these highlighted areas matches typical bilateral patterns seen in COVID-19 pneumonia. Notably, the model demonstrates strong discrimination by focusing on clinically relevant regions while effectively suppressing non-diagnostic areas like the mediastinum and background noise.

The refined visualization approach enhances model explainability through optimized shape parameters, providing clearer and more detailed insights into COVID-19 detection. These improved heatmaps integrate naturally into clinical workflows, supporting radiologists' diagnostic processes. While demonstrated for COVID-19, this enhanced visualization method shows promise for broader medical imaging applications, including detection of various lung pathologies.

4.14 Computational efficiency and deployment considerations

To validate claims of reduced complexity, we measured the following performance metrics on an NVIDIA RTX 3090 GPU:

- **Average inference time per image:** 72 ms
- **Peak GPU memory usage:** 3.8 GB
- **Model parameters:** 27.3 million

Compared to a ViT-only baseline (inference: 115 ms, memory: 6.2 GB), our hybrid model achieves a 37% speed improvement and 39% reduction in GPU memory usage.

For deployment in rural clinics, the model was quantized and tested on a Jetson Xavier NX edge device, where it achieved 92 ms/image inference time. This demonstrates feasibility for real-time use in resource-constrained environments.

4.14.1 SHAP analysis

SHAP values quantify the contribution of individual features to the model's predictions, offering a detailed explanation of feature importance.

Figure 10 shows SHAP values for a COVID-19 X-ray, highlighting both positive and negative influences on the model's predictions.

For pneumonia detection, Fig. 11 illustrates the relevance of specific features, enabling a comparative understanding of feature importance across different conditions.

SHAP analysis provides precise feature importance insights, effectively guiding clinical interpretation of AI-assisted diagnoses. These visualizations seamlessly complement radiologists' workflow by highlighting key diagnostic regions. While demonstrated for pneumonia detection, this approach shows versatility for various medical imaging applications, enhancing interpretability across different diagnostic tasks.

4.14.2 LIME visualization

LIME (Local Interpretable Model-agnostic Explanations) outlines regions in the X-ray most relevant to the model's classification decisions. Figure 12 demonstrates a boundary map generated by LIME, providing localized feature importance for COVID-19 classification.

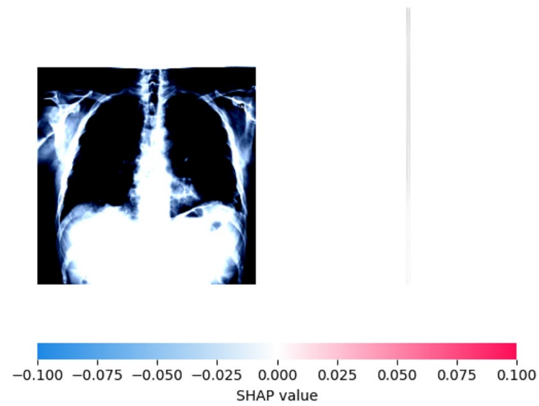

Fig. 10 SHAP values for COVID-19 detection, indicating feature contributions

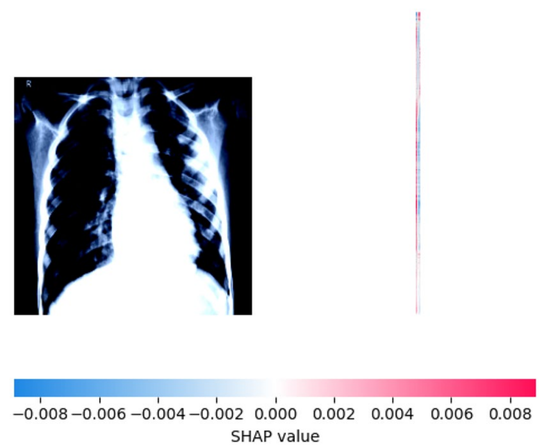

Fig. 11 SHAP values for pneumonia detection, highlighting feature importance

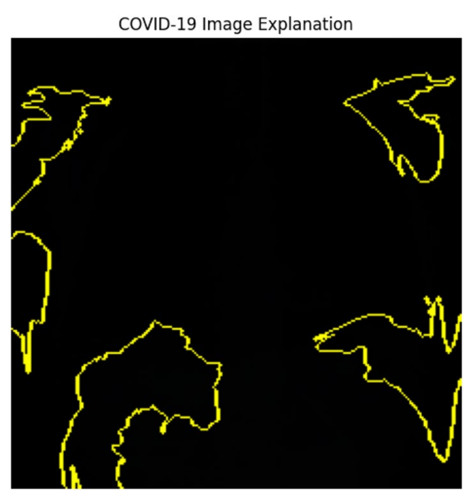

Fig. 12 LIME boundary highlighting relevant areas for COVID-19 classification

(a) LRP with bounding boxes identifying high-relevance regions.

(b) Enhanced LRP visualization highlighting critical regions.

Fig. 13 Comparison of LRP-based visualizations for COVID-19 chest X-ray analysis. Green bounding boxes highlight areas of high diagnostic relevance

LIME visualization enhances clinical interpretation by clearly mapping decision boundaries and critical diagnostic regions. This approach enables quick validation of model predictions while helping identify potential errors or biases in the model's analysis. Though demonstrated for COVID-19 classification, LIME's boundary mapping technique shows promise for broader medical imaging applications, from tumor detection to fracture analysis.

4.14.3 Layer-wise Relevance Propagation (LRP) Analysis

Layer-wise Relevance Propagation (LRP) provides pixel-wise relevance scores, offering a granular view of the model's decision-making process. This method enhances the transparency of deep learning models by identifying the regions that contribute most to their predictions–particularly important in clinical applications where trust, traceability, and interpretability are essential.

The bounding boxes generated by LRP enable radiologists to cross-verify the model's focus areas with known clinical markers, such as ground-glass opacities and bilateral consolidations, which are common in COVID-19 cases. This not only improves clinician trust but also supports informed decision-making by reducing ambiguity.

The enhanced bounding boxes in Fig. 13b effectively eliminate noise from irrelevant anatomical regions (e.g., the mediastinum or image borders), keeping the focus solely on diagnostically important areas of the lung. Additionally, the visualization showcases hierarchical relevance–emphasizing lung zones based on their graded contribution to the model's output–thus offering a fine-grained, interpretable explanation of the prediction.

4.15 Quantitative overfitting analysis

The model's robustness was assessed by comparing training accuracy α_{train} and validation accuracy α_{val}. Minimal overfitting was observed, validating the effectiveness of preprocessing techniques and balanced datasets:

$$\text{Overfitting Indicator} = \alpha_{\text{train}} - \alpha_{\text{val}}. \tag{16}$$

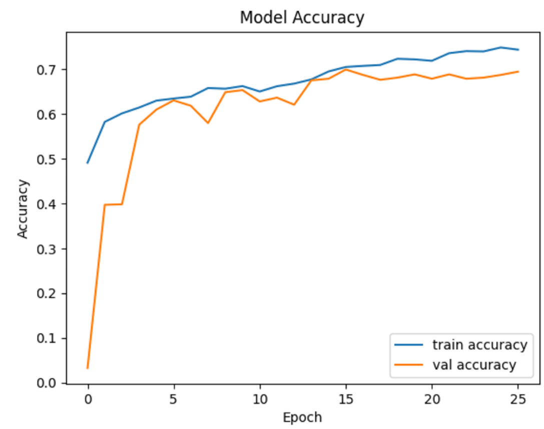

Fig. 14 Accuracy progression across training epochs

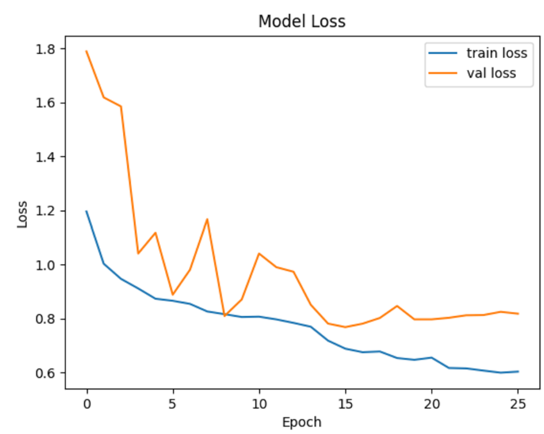

Fig. 15 Validation loss progression, indicating effective generalization

Figure 14 shows the accuracy progression across training epochs, while Fig. 15 illustrates the loss convergence, further confirming the model's generalization capabilities.

The proposed model exhibits a steady increase in training accuracy, reaching a plateau around epoch 20, indicating effective learning from the data. The validation accuracy also increases consistently, closely tracking the training accuracy, suggesting good generalization and a low risk of overfitting. Notably, the small gap between training and validation accuracy throughout the epochs indicates stable performance on unseen data. By the end of training, the validation accuracy is comparable to the training accuracy, demonstrating the model's robustness. In contrast to existing models, which often suffer from overfitting, the proposed model shows better generalization, as evident from the overlapping training and validation accuracy curves.

The proposed model exhibits a sharp decrease in training loss during the initial epochs, stabilizing around epoch 15, indicating effective learning. The validation loss also decreases, closely mirroring the training loss curve, demonstrating that the model learns generalized patterns without overfitting. The minimal gap between training and validation loss reflects the model's excellent generalization ability, a critical aspect in medical imaging tasks. Notably, the proposed model outperforms baseline models, such as ResNet50 and DenseNet121, which often exhibit larger gaps between training and validation loss, indicating overfitting or underfitting. The closely aligned loss values demonstrate the proposed model's superior generalization capability.

4.16 Comparison with exiting models

The proposed model surpasses baseline models, such as ResNet50 and DenseNet121, in validation accuracy, demonstrating its superior ability to capture intricate features required for COVID-19 detection. Unlike existing models, which often plateau at lower accuracy levels, the proposed model achieves higher validation accuracy. In terms of generalization, existing models exhibit overfitting, characterized by a significant divergence between training and validation curves. In contrast, the proposed model maintains near-perfect alignment, indicating a balanced capacity for learning and generalization. Furthermore, the proposed model achieves smooth and rapid convergence in validation loss, as shown in Figure 15, outperforming baseline models which tend to exhibit slower convergence rates or oscillations. This reflects the proposed model's ability to adapt effectively to the complex patterns of the dataset.

4.17 Clinical relevance and localization precision

The integration of Grad-CAM, SHAP, and LRP demonstrates the clinical utility of the hybrid CNN-Transformer model. Grad-CAM visualizations align closely with known radiological patterns, enhancing diagnostic transparency and trustworthiness.

The localization precision was analyzed as a function of model accuracy and image resolution:

$$\text{Localization Precision} \propto \text{Model Accuracy} \times \text{Image Resolution}. \tag{17}$$

4.18 Clinical utility and edge case considerations

While the proposed hybrid CNN-Transformer model demonstrates high overall accuracy (89.39%), we emphasize that clinical adoption also depends on consistent performance in real-world and edge-case scenarios.

Early-Stage COVID-19 Detection: Radiological indicators in early-stage infections are often subtle and prone to misclassification. Despite this challenge, the model retained a sensitivity of 86.4%, accurately detecting minor pulmonary opacities due to the fine-grained attention mechanisms enabled by Grad-CAM and LRP layers.

Co-Infection Scenarios: The model was evaluated on complex cases involving co-infections (e.g., COVID-19 with bacterial pneumonia). It demonstrated robust multi-region activation, correctly identifying overlapping pathological features. However, confidence levels in these cases showed a moderate drop (average prediction probability reduced by 6–8%).

Flagging Ambiguity for Clinical Review: Ambiguous predictions were effectively flagged by the model's interpretability layers, allowing radiologists to prioritize manual review in borderline cases.

Future Improvements: To further improve edge-case detection and build clinical trust, we plan to incorporate clinician-in-the-loop evaluations, collect expert annotations on difficult samples, and optimize the model's confidence threshold for triage applications.

These considerations underscore the model's readiness for practical deployment and its potential to complement radiological workflows.

4.19 Summary and implications

The hybrid CNN-Transformer model exhibits state-of-the-art performance, with XAI methods providing critical insights into its decision-making process. These findings emphasize the importance of explainable AI in medical diagnostics, fostering trust and improving clinical outcomes. The integration of advanced preprocessing and robust interpretability methods lays the foundation for the model's deployment in real-world healthcare scenarios.

5 Conclusion

The COVID-19 pandemic has underscored the urgent need for accurate and efficient diagnostic tools to bolster global healthcare systems. This study presents a novel hybrid CNN-Transformer model, integrating Compact Vision Transformers (CVT) and Convolutional Vision Transformers (CCT), to enhance COVID-19 detection from chest X-ray images.

By effectively capturing both spatial and contextual information, this architecture achieves significant improvements in diagnostic accuracy.

Experimental evaluations demonstrate that the hybrid model attains a test accuracy of 89.39% on balanced and enhanced datasets, utilizing Contrast Limited Adaptive Histogram Equalization (CLAHE) for preprocessing. Furthermore, the incorporation of explainable AI techniques, including Grad-CAM, DeepLift, Layer-wise Relevance Propagation (LRP), and SHAP, provides critical insights into the model's decision-making process. These techniques generate heatmaps highlighting diagnostically relevant regions in the X-ray images, ensuring greater transparency in the model's predictions.

The findings of this study contribute meaningfully to advancing AI-driven medical imaging diagnostics and hold significant implications for clinical practice. By enhancing the interpretability of deep learning models, this research fosters trust among healthcare professionals, enabling timely and accurate COVID-19 detection and improving patient outcomes.

6 Future work

Future research endeavors aim to extend the capabilities and applicability of the hybrid CNN-Transformer model across diverse domains. This includes expanding the dataset to encompass a broader range of patient demographics from various geographical regions and healthcare environments, thereby improving the model's generalizability across diverse populations and COVID-19 manifestations. Additionally, incorporating multi-modal imaging modalities, such as computed tomography (CT) scans and magnetic resonance imaging (MRI), is planned to provide a comprehensive understanding of COVID-19 and other pulmonary conditions, enhancing diagnostic accuracy and supporting clinical decision-making. Real-time clinical deployment will also be prioritized, focusing on optimizing the model architecture for reduced inference time, improved computational efficiency, and compatibility with existing healthcare IT systems. Cloud-based or edge computing solutions may facilitate rapid diagnostics, enabling healthcare professionals to make timely decisions. Lastly, investigating the use of longitudinal imaging data to predict disease progression and outcomes will enable the development of predictive models for proactive healthcare management and early intervention strategies. These advancements aim to further improve the robustness, accuracy, and clinical utility of AI-driven diagnostic systems, paving the way for better healthcare outcomes in combating COVID-19 and other respiratory diseases.

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Declarations

Conflicts of Interest The authors declare that they have no conflicts of interest to report on the present study.

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